

4. In a class experiment, Dylan and Joel were given a solution of sodium carbonate containing 5.23 g of sodium carbonate powder dissolved in 500 cm³ of water.

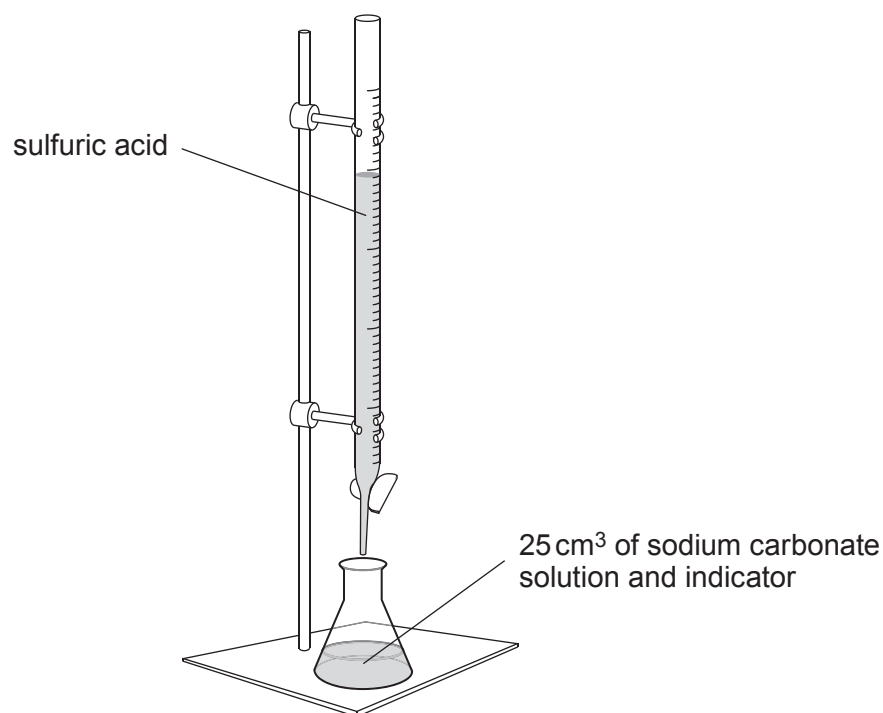
The relative formula mass (M_r) of sodium carbonate is 106.

- (a) Use this information to calculate the number of moles of sodium carbonate in the 5.23 g of the powder. Give your answer correct to **two** decimal places. [2]

Number of moles = mol

- (b) Dylan and Joel were asked to use the sodium carbonate solution to prepare a sample of sodium sulfate crystals in a three-stage process.

In the first stage of their preparation, they used the following apparatus to carry out a titration.



The equation for the reaction taking place is as follows.



- (i) A trial run was carried out and the titration repeated three times. The volume of acid added each time was recorded.

	Trial	1	2	3
Volume of sulfuric acid added (cm ³)	30.20	27.55	27.75	27.65

- I. State the purpose of carrying out a trial run.

[1]

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- II. State whether the sulfuric acid or the sodium carbonate solution is the more concentrated. Give the reason for your answer.

[1]

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- (ii) Use **all** of the information provided to describe **in detail** the other two stages Dylan and Joel carried out to obtain pure sodium sulfate crystals.

[4]

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- (c) Sodium sulfate solution is also formed when sodium hydroxide solution reacts with copper(II) sulfate solution.

(i) Give the balanced symbol equation for this reaction. [3]

(ii) Describe tests that can be carried out to identify both of the ions in sodium sulfate solution. Give the expected observation for both tests. [2]

- (d) The reaction occurring between solutions of sodium carbonate and magnesium sulfate forms a precipitate of magnesium carbonate.

Write the **ionic** equation for the formation of magnesium carbonate. [2]

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